Synaptic State or KV-Cache? A Unified Associative-Memory View, a Dual-Timescale Architecture Class, and a Matched-Compute Protocol

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Abstract

Softmax attention stores context in a key–value cache that grows with sequence length; Hebbian "synaptic" models such as the Dragon Hatchling (BDH) store it in a fixed-size, positive, high-dimensional state rewritten at every token; and recurrent latent reasoners such as BDH-CQ write demonstrations into such a state and then iterate in latent space. We ask *when, and at what cost, inference-time memory should be synaptic rather than a cache.* We (i) cast all three families as one associative-memory operator with explicit write, read, forgetting, consolidation, and workspace maps, with byte accounting that yields a closed-form memory crossover $T^\star$; (ii) prove that any state with data-independent exponential forgetting couples retention and interference, so no decay rate can both adapt quickly and retain earlier episodes; (iii) specify *Compact Dual-Timescale Latent Reasoning* (CDLR), an architecture class pairing a fast episodic Hebbian state with a slow, gated, consolidated bank, a halting latent workspace, a local readout adapter, and an audit trail; (iv) audit the empirical claims of BDH and BDH-CQ, separating limitations the authors state from gaps we infer; and (v) give a preregistered evaluation protocol, matched on parameters, tokens, and compute, that reports accuracy, memory, and latency jointly. We report no large-scale results: the contribution is the framework, the analysis, four falsifiable hypotheses, and the protocol needed to test them. A CPU pilot of the capacity track, at toy widths $n \leq 4096$ with synthetic keys and no trained encoder, only characterises regimes: a raw positive Hebbian state plateaus; centred Hebbian and delta-rule states with random keys give fitted exponents 0.86–0.94; and keys whose shared structure the write rule cannot remove (fixed-correlation keys under the raw rule, strongly correlated keys under the delta rule, or clustered keys) give flat capacity, a different regime from the $\widetilde{\Omega}(\sqrt{n})$ worst case.

1 Introduction

The Transformer (Vaswani et al., 2017) accepts an $O(T^2 d)$ per-layer cost in sequence length T in exchange for constant path length and full parallelism (Vaswani et al., 2017, §4, Table 1). At inference it keeps every past key and value: context is *stored*, not *learned*. *The Dragon Hatchling* (BDH; Kosowski et al., 2025) argues instead that inference-time working memory can rely "entirely on synaptic plasticity with Hebbian learning" (Abstract), in a per-layer state of size $n \times d$ that does not grow with context (§3, Eq. 5), and reports parity with GPT-2-style baselines from 10M to 1B parameters on Europarl translation (§4.2, Fig. 7). *BDH-CQ* (Engdahl et al., 2026) writes demonstrations into a recurrent memory and solves a query by iterating in latent space (§3.2–3.3, Eqs. 1–4), reporting 29.5% pass@2 on the public ARC-AGI-1 evaluation at \$0.00070 per task (§5, Table 1).

These claims leave open the question that matters for the next architecture class: **is a fixed-size synaptic state a better inference-time memory than a KV-cache, and under what accounting?** Three observations motivate this paper. First, the families are described in incompatible vocabularies (operators, synapses, recurrent memory), which hides what actually differs: what is written, how it is forgotten, and how it is read. Second, "bounded" is not "small": a BDH state with $n = 32{,}768$, $d = 256$ (Kosowski et al.,

2025, §4.1) is larger than a 768-wide Transformer's KV-cache until several thousand tokens (Figure 1). Third, a single decaying state cannot both adapt to a new task and keep an old one (Proposition 1); BDH's merged-model language mixing (Kosowski et al., 2025, §7.1) is consistent with this, and complementary learning systems (McClelland et al., 1995; Kumaran et al., 2016) and synaptic consolidation models (Fusi et al., 2005; Benna & Fusi, 2016) suggest the remedy of a second, slower timescale. Meanwhile, the evidence base is narrow: neither paper compares against modern linear-recurrent models such as Mamba (Gu & Dao, 2023), RetNet (Sun et al., 2023), xLSTM (Beck et al., 2024), gated/delta linear attention (Yang et al., 2024a;b; 2025), or Titans (Behrouz et al., 2025) at matched compute.

Contributions. None of the contributions below depends on a new large-scale experiment.

C1. Unified formalisation (Section 3). An associative-memory operator $(\mathcal{M}, \omega, \rho, \lambda, \kappa, \mathcal{F})$ covering softmax attention, linear attention and SSMs, BDH-GPU, graph-form BDH, BDH-CQ, and CDLR, with byte accounting and a closed-form crossover $T^\star$.

C2. Retention–interference coupling (Proposition 1). For data-independent exponential forgetting, retention of an old episode and its interference with a new one are the same function of lag.

C3. The CDLR architecture class (Section 4), with hypotheses H1–H4 stated so that each can be refuted.

C4. A source-anchored audit of BDH and BDH-CQ (Section 6), separating paper-stated limitations$^{(\mathrm{P})}$ from inferred gaps$^{(\mathrm{I})}$.

C5. A preregistered matched-compute protocol (Section 5) with five tracks E1–E5, an ablation ladder with a byte-transfer control (Figure 4), Pareto reporting, and a negative-results policy.

C6. A toy capacity pilot of E2 (Figure 3, Section A P5), labelled PILOT. A regime characterisation of the memory rule alone at $n \leq 4096$. It is supporting evidence for the protocol's design, not a test of H1 or of BDH Claim 7.

Scope. This is a framework, analysis, and protocol paper. We do *not* claim ARC-AGI parity with BDH-CQ, scaling to hundreds of billions of parameters, or a solution to the no-backpropagation-through-time gap of Kosowski et al. (2025, §7.2). Small CPU smoke tests of our code are confined to Section A and support no claim in the main text. The one empirical statement we make in the main text is the E2 PILOT (Figure 3). It is scoped to *regime characterisation at toy width* $n \leq 4096$ ($d = 32$, synthetic k-WTA keys, closed-set recall, 3 seeds). We do **not** claim (i) general $\tilde{O}(n)$ or $\tilde{\Omega}(\sqrt{n})$ capacity for BDH or any trained model, (ii) any result of a Transformer-matched (parameter-, token-, or FLOP-matched) bakeoff, or (iii) anything about the behaviour of states driven by trained encoders.

2 Background and Related Work

Softmax attention computes $\mathrm{softmax}(QK^\top/\sqrt{d_k})V$ (Vaswani et al., 2017, Eq. 1). Later work made it IO-efficient (Dao et al., 2022) and added relative positions (Su et al., 2024; Press et al., 2022) and segment recurrence (Dai et al., 2019). Its inference-time KV-cache, which Kosowski et al. (2025, Table 3) describe as "not localized at neurons", grows linearly in T.

Linear attention, fast weights, and SSMs. With a feature map ϕ, attention becomes a recurrence on $\sum_\tau \phi(k_\tau) v_\tau^\top$ (Katharopoulos et al., 2020; Choromanski et al., 2021), a fast-weight programmer (Ba et al., 2016; Schlag et al., 2021). Gated, decayed, and selective variants include RetNet (Sun et al., 2023), GLA (Yang et al., 2024a), Mamba (Gu & Dao, 2023), Mamba-2 (Dao & Gu, 2024), xLSTM (Beck et al., 2024), DeltaNet (Yang et al., 2024b), and Gated DeltaNet (Yang et al., 2025). Test-time training (Sun et al., 2024), Titans (Behrouz et al., 2025), and test-time regression (Wang et al., 2025b) read these as online optimisation of an associative memory. We follow this line and add two things: BDH's positive high-dimensional neuron state and BDH-CQ's latent workspace in the same frame, and an explicit *consolidation* map κ, which prior unifications leave implicit or absent. Fixed-state models trail attention on associative recall and copying

(Arora et al., 2023; Jelassi et al., 2024), so any claim for synaptic memory must be tested on exactly those tasks (E2).

BDH models a language model as local graph dynamics among n neurons with Hebbian synapse reweighting (Kosowski et al., 2025, §2). Its tensor form BDH-GPU(n, d) has $(3 + o(1))nd$ parameters, positive ReLU activations, and linear attention in neuron dimension n (§3, Def. 4, Eq. 4); it scales in n with $d = 256$ (App. B.2). The authors report ∼5% non-zero activations (§4.1), modular scale-free structure (§5), and monosemantic synapses (§6.3). Claim 7 (§6.1, App. C.2) states that linear attention in $\mathbb{R}^n$ can separate $\tilde{O}(n)$ facts under non-adversarial keys and $\tilde{\Omega}(\sqrt{n})$ without assumptions.

BDH-CQ and latent reasoning. BDH-CQ writes demonstrations as $\mathbf{S}_t = U_\theta(\mathbf{S}_{t-1}, D_t)$, initialises $\mathbf{H}_0 = E_\theta(x^*, \mathbf{S}_K)$, iterates $\mathbf{H}_{r+1} = F_\theta(\mathbf{H}_r, \mathbf{S}_K)$, and decodes $\hat{y} = G_\theta(\mathbf{H}_R)$ (Engdahl et al., 2026, Eqs. 1–4). Related recurrent-depth approaches include Universal Transformers (Dehghani et al., 2019), ACT (Graves, 2016), Coconut (Hao et al., 2024), and recurrent-depth scaling (Geiping et al., 2025), in contrast to verbal chain of thought (Wei et al., 2022). On ARC (Chollet, 2019), HRM (Wang et al., 2025a) and TRM (Jolicoeur-Martineau, 2025) are recurrent solvers.

Continual learning and complementary learning systems. Sequential training causes catastrophic interference (McCloskey & Cohen, 1989; French, 1999); remedies include regularisation (Kirkpatrick et al., 2017; Zenke et al., 2017), replay (Lopez-Paz & Ranzato, 2017; Rolnick et al., 2019), and episodic control (Pritzel et al., 2017), and long streams erode plasticity (Dohare et al., 2024). CLS theory holds that a fast hippocampal system stores episodes and a slow neocortical system consolidates them (McClelland et al., 1995; Kumaran et al., 2016); cascade synapse models extend memory lifetime by coupling time constants (Fusi et al., 2005; Benna & Fusi, 2016). Hebbian and three-factor rules (Hebb, 1949; Oja, 1982; Gerstner et al., 2018) and differentiable plasticity (Miconi et al., 2018) supply our local updates. CDLR differs in placing the slow timescale *at inference, across episodes*, with explicit gating, a fixed byte budget, and an audit trail.

3 A Unified Formalisation of Inference-Time Memory

Notation. T is sequence length and n the BDH neuron count; both are often written "n" (Vaswani et al., 2017, Table 1, vs. Kosowski et al., 2025, §3), which produces misleading complexity comparisons. d is the low-rank width, L the number of layers, b bytes per stored scalar (full notation in Section C).

Definition 1 (Associative-memory operator)**.** *An inference-time memory layer is a tuple* $(\mathcal{M}, \omega, \rho, \lambda, \kappa, \mathcal{F})$*: a state space* $\mathcal{M}$*; a* write $\omega : \mathcal{M} \times \mathcal{X} \to \mathcal{M}$*; a* read $\rho : \mathcal{M} \times \mathcal{Q} \to \mathbb{R}^{d_v}$*; a* forgetting *map* λ *(decay, gating, or eviction); an optional* consolidation *map* $\kappa : \mathcal{M} \times \mathcal{B} \to \mathcal{B}$ *into a slower store* $\mathcal{B}$*; and an optional* workspace $\mathcal{F}$ *that iterates reads before emitting output. Slow parameters* θ *are fixed at inference unless stated.*

Table 1 instantiates the definition. Rows differ in (a) whether $\mathcal{M}$ grows with T; (b) whether keys live in a low signed space ($d_k = 64$) or a high positive sparse space ($n \sim 10^4$–10^6); (c) whether κ and $\mathcal{F}$ exist. Softmax attention is a *nonparametric* memory (a kernel smoother over stored pairs). Linear attention, SSMs, and BDH-GPU are *parametric* memories whose read matrix is learned online by a Hebbian or delta rule (Schlag et al., 2021); BDH-GPU adds positive sparse keys with $n \gg d$ and cross-layer weight sharing (Kosowski et al., 2025, §4.1). BDH-CQ adds a workspace $\mathcal{F}$ that re-reads a fixed $\mathbf{S}_K$ for R steps (Engdahl et al., 2026, Eq. 3), separating iterative computation from token emission.

Byte accounting. Per layer and sequence, a softmax KV-cache stores $2Td_{\text{model}}b$ bytes; BDH-GPU stores $nd\,b$ bytes regardless of T (Kosowski et al., 2025, Eq. 5). They are equal at

$$T^\star = \frac{n\,d}{2\,d_{\text{model}}}. \tag{1}$$

For the 25M BDH-GPU configuration ($n = 32{,}768$, $d = 256$; Kosowski et al., 2025, §4.1) against $d_{\text{model}} = 768$, $T^\star \approx 5{,}461$ tokens per layer$^{(\mathrm{I})}$ (Figure 1). Below $T^\star$ the bounded state is the *larger* memory. This is consistent with, and independent of, the paper's FLOP crossover near 4096 tokens at $d = 256$ (§4.2). Constant memory

Table 1: Inference-time memory as an associative-memory operator (Definition 1). "n.d." = not disclosed. ϕ is a feature map, $\odot$ the Hadamard product, U a decay/rotation operator, G_s a sparse synapse mask.

Model	State $\mathcal{M}$ and write ω	Read ρ	Forget λ	κ	$\mathcal{F}$
Softmax attn. (Vaswani et al., 2017)	append (k_t, v_t); size $2Td$	$\mathrm{softmax}(q^\top K/\sqrt{d_k})V$	none (window)	–	–
Linear attn. (Katharopoulos et al., 2020)	$\mathbf{S} \leftarrow \mathbf{S} + \phi(k)v^\top$; $d_\phi \times d$	$\phi(q)^\top \mathbf{S} / \phi(q)^\top z$	none	–	–
RetNet/GLA/Mamba-2 (Sun et al., 2023; Yang et al., 2024a; Dao & Gu, 2024)	$\mathbf{S} \leftarrow \Lambda_t \mathbf{S} + kv^\top$	$q^\top \mathbf{S}$	fixed or input-gated	–	–
DeltaNet (Yang et al., 2024b; 2025)	$\mathbf{S} \leftarrow \mathbf{S} + \beta k(v - \mathbf{S}^\top k)^\top$	$q^\top \mathbf{S}$	overwrite (+gate)	–	–
Titans (Behrouz et al., 2025)	MLP weights by surprise gradient	MLP(q)	weight decay	–	–
BDH-GPU (Kosowski et al., 2025, Eq. 4–5)	$\rho \leftarrow (\rho + v^* x^\top)U$; $n \times d$/layer, $x \geq 0$ sparse	$\rho\, x_t$ (positive $x_t \in \mathbb{R}^n$)	RoPE/ALiBi-like U	–	–
BDH graph (Kosowski et al., 2025, Eq. 6)	$\sigma \leftarrow \sigma + (yx^\top \odot G_s)U$ on edges	local E/I dynamics	U	–	–
BDH-CQ (Engdahl et al., 2026, Eqs. 1–4)	$\mathbf{S}_t = U_\theta(\mathbf{S}_{t-1}, D_t)$, n.d.	via E_θ, F_θ, n.d.	n.d.	–	R loops
CDLR (this work)	fast $\mathbf{S}$ (Hebbian/delta) + slow bank $\mathbf{C}$	normalised read of $\mathbf{S}$, gated $\mathbf{C}$	decay on $\mathbf{S}$; gated on $\mathbf{C}$	✓	halting R

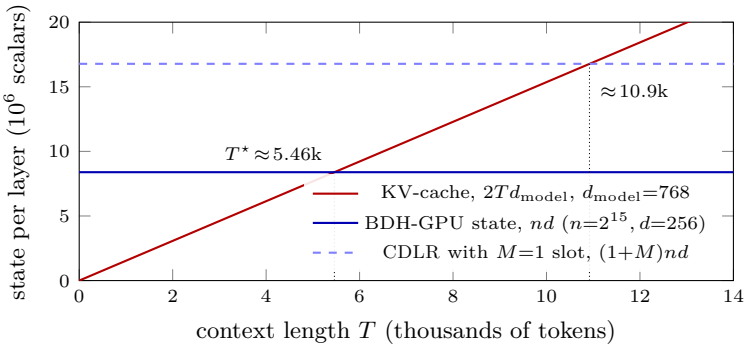


Figure 1: Conceptual byte crossover from Equation (1). Lines are the closed-form state sizes, not measurements. Each consolidated slot moves the crossover right by $T^\star$; the residual buffer $\mathcal{R}$ and parameter bytes are omitted.

pays off only for long contexts or many concurrent episodes, so fair comparisons must report bytes at several T, not asymptotic class alone.

Capacity. Interference in a linear associative state grows with key overlap. BDH Claim 7 gives $\tilde{O}(n)$ separable facts for non-adversarial keys and $\tilde{\Omega}(\sqrt{n})$ in the worst case; to our knowledge it has not been measured on recall benchmarks (Arora et al., 2023). This becomes H1 and E2. A toy PILOT (Figure 3) suggests that which regime appears depends on the write rule and on how key correlation is parameterised, and not only on n.

3.1 Single-timescale states couple retention and interference

Consider $\mathbf{S}_t = \lambda \mathbf{S}_{t-1} + u_t$, $0 < \lambda < 1$, with write u_t (e.g. Hebbian $\phi(k_t)v_t^\top$). A stream switches at t_0 from task A (writes u_t^A) to task B (writes u_t^B), both stationary in mean with means $\bar{u}^A, \bar{u}^B$.

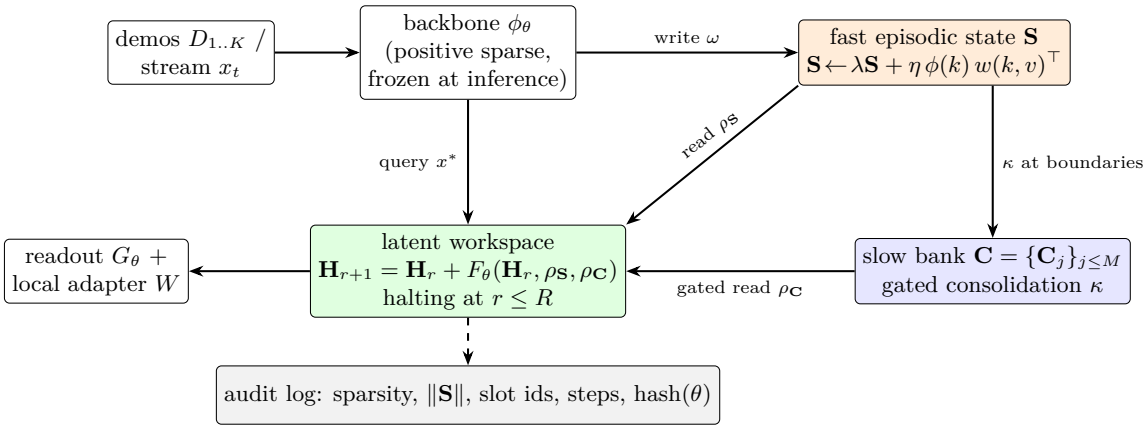


Figure 2: CDLR data flow. Orange: fast timescale (within episode, decays). Blue: slow timescale (across episodes, gated, not overwritten by an unmatched task). Green: latent workspace. Slow parameters θ are frozen at inference; only $\mathbf{S}$, $\mathbf{C}$, and the optional adapter change, and the audit log records every change.

Proposition 1 (Retention–interference coupling)**.** *At lag $\Delta \ge 0$ after the switch, in expectation and in the stationary limit before t_0,*

$$\mathbb{E}\,\mathbf{S}_{t_0+\Delta} = \tfrac{1}{1-\lambda}\big(\lambda^\Delta\, \bar{u}^A + (1-\lambda^\Delta)\, \bar{u}^B\big).$$

The normalised weight of task A, $\alpha(\Delta) = \lambda^\Delta$, is simultaneously (i) the retention *of A and (ii) the* interference *of A with B-queries. With retention horizon $H_{\mathrm{ret}}(\varepsilon) = \min\{\Delta : \alpha(\Delta) \le \varepsilon\}$ and adaptation horizon $H_{\mathrm{ad}}(\delta) = \min\{\Delta : \alpha(\Delta) \le \delta\}$, for any λ, $H_{\mathrm{ret}}(\varepsilon)/H_{\mathrm{ad}}(\delta) \to \log\varepsilon/\log\delta$, independent of λ. In particular, if A must stay retrievable ($\alpha \ge \varepsilon$) and B uncontaminated ($\alpha \le \delta$) at the same lag, with $\varepsilon > \delta$, no λ satisfies both.*

Proof. Unrolling, $\mathbf{S}_{t_0+\Delta} = \lambda^\Delta \mathbf{S}_{t_0} + \sum_{j=1}^{\Delta} \lambda^{\Delta-j} u^B_{t_0+j}$. In the stationary limit $\mathbb{E}\mathbf{S}_{t_0} = \bar{u}^A/(1-\lambda)$, and $\sum_{j=1}^{\Delta} \lambda^{\Delta-j} = (1-\lambda^\Delta)/(1-\lambda)$, which gives the expression. Both horizons solve $\lambda^\Delta = c$, i.e. $\Delta = \lceil \log c/\log\lambda \rceil$, so their ratio tends to $\log\varepsilon/\log\delta$ (exactly, up to rounding), independent of λ. Requiring $\lambda^\Delta \ge \varepsilon > \delta \ge \lambda^\Delta$ is a contradiction. □

Remark 1 (Scope). The statement concerns one shared, linearly superposed state. It does not rule out good continual behaviour for (a) input-dependent gates (λ_t in Mamba/GLA/Gated DeltaNet), which can freeze the state but then also stop adapting unless the gate infers task boundaries; (b) delta rules, which remove interference along overwritten keys but erase the overwritten association; (c) high-dimensional sparse keys, which reduce overlap between $\bar{u}^A$ and $\bar{u}^B$. Case (c) is BDH's implicit bet; it holds only while tasks occupy near-disjoint neurons, which the language mixing of Kosowski et al. (2025, §7.1) suggests is not guaranteed. Cascade models (Fusi et al., 2005; Benna & Fusi, 2016) escape the coupling by adding variables with different time constants; this is the design principle behind CDLR.

4 Compact Dual-Timescale Latent Reasoning (CDLR)

CDLR is an architecture *class*: the operator of Definition 1 with both κ and $\mathcal{F}$ present and a hard byte budget (Figure 2). Until E3 (with its byte-transfer control) and E4 are run, it is a design and evaluation target, not a demonstrated method.

Backbone. $\phi_\theta : \mathcal{X} \to \mathbb{R}^n_{\ge 0}$ produces positive, sparse, high-dimensional codes. The reference instantiation is a BDH-GPU stack (Kosowski et al., 2025, Def. 4) with k-winners-take-all and ℓ_2 normalisation. A Transformer or SSM backbone with a positive projection head is a control that separates backbone effects from memory-architecture effects.

Fast episodic state. $\mathbf{S} \in \mathbb{R}^{n \times d_v}$ with normaliser $z \in \mathbb{R}^n$:

$$\mathbf{S} \leftarrow \lambda \mathbf{S} + \eta\, \phi(k)\, w_\theta(k, v)^\top, \qquad z \leftarrow \lambda z + \eta\, \phi(k), \qquad \rho_{\mathbf{S}}(q) = \frac{\phi(q)^\top \mathbf{S}}{\phi(q)^\top z + \epsilon}, \tag{2}$$

with $\eta = \mathrm{softplus}(\cdot)$ and $\lambda = \mathrm{sigmoid}(\cdot)$ slow parameters. The delta-rule variant $\mathbf{S} \leftarrow \lambda \mathbf{S} + \eta\, \phi(k)(w - \mathbf{S}^\top \phi(k))^\top$ (Schlag et al., 2021; Yang et al., 2024b) is a first-class ablation because it changes the capacity regime of H1.

Slow consolidated bank and consolidation κ. $\mathbf{C} = \{(\mathbf{C}_j, \mathbf{k}_j)\}_{j=1}^M$ holds at most M slots shaped like $\mathbf{S}$ (or rank-r factorised), each with a summary key $\mathbf{k}_j$, plus a small reservoir $\mathcal{R}$ of past codes for replay (McClelland et al., 1995). At a consolidation event (episode boundary, or a novelty trigger when the read error of $\mathbf{S}$ exceeds a threshold) we form $\hat{\mathbf{C}} = \mathrm{unit}\big(\frac{1}{2}\mathbf{S} + \frac{1}{2}\,\Pi(\mathcal{R})\big)$, where Π computes key-conditioned means of replayed codes, and let $j^\star = \arg\max_j \cos(\hat{\mathbf{C}}, \mathbf{C}_j)$:

$$\kappa: \begin{cases} \mathbf{C}_{j^\star} \leftarrow (1-\beta)\mathbf{C}_{j^\star} + \beta \hat{\mathbf{C}} & \text{if } \cos(\hat{\mathbf{C}}, \mathbf{C}_{j^\star}) \geq \tau, \\ \text{allocate a new slot (evict least-used if } |\mathbf{C}| = M) & \text{otherwise.} \end{cases} \tag{3}$$

Reads are gated, $g_j = \mathrm{softmax}_j(\gamma s_j)$, with s_j either key similarity $\cos(\mathbf{k}_j, \bar{\phi}_{\mathrm{ctx}})$ (unsupervised) or the fit of $\mathbf{C}_j$ to a labelled support set (supervised; this leaks task information and must be reported as such); the term is dropped when $\max_j s_j < \tau_{\mathrm{gate}}$. Because $\mathbf{C}_j$ is written only when matched, task A's slot is not diluted by task B's writes. This breaks the coupling of Proposition 1 at the cost of M extra states and a task-boundary or novelty signal.

Latent workspace. Following BDH-CQ's public equations (Engdahl et al., 2026, Eqs. 2–4) with our own open choices for every function:

$$\mathbf{H}_0 = E_\theta\big(x^*, \rho_{\mathbf{S}}(\phi(x^*)), \textstyle\sum_j g_j \rho_{\mathbf{C}_j}(\phi(x^*))\big), \qquad \mathbf{H}_{r+1} = \mathbf{H}_r + F_\theta\big([\mathbf{H}_r; \rho_{\mathbf{S}}(\kappa_k(W_q \mathbf{H}_r))]\big),\ r < R, \tag{4}$$

$$\hat{y} = G_\theta(\mathbf{H}_{r^\star}), \quad r^\star = \min\{r : \max \mathrm{softmax}(G_\theta(\mathbf{H}_r)) \geq c\} \wedge R, \tag{5}$$

where κ_k is k-WTA and c a halting threshold as in ACT (Graves, 2016). Every re-read uses the same $\mathbf{S}$, so the workspace adds compute but no memory that grows with R.

Local adapter and audit. An optional readout $W\phi + b$ adapts online by a three-factor rule (Gerstner et al., 2018), $e_t = \gamma e_{t-1} + \phi_t$, $W \mathrel{+}= \eta_W (y_t - p_t) e_t^\top$; nothing flows into the backbone or back through time. For a linear head with $\gamma = 0$ this is the cross-entropy gradient of that head, so its locality lies in what it does *not* touch; we do not claim it closes the no-BPTT gap of Kosowski et al. (2025, §7.2). Every `predict` call emits activation sparsity, $\|\mathbf{S}\|_F$, $\|\Delta\mathbf{H}\|$ per step, halting step, selected slots and gates, consolidation actions, latency, and a hash of θ before and after, turning "no parameter updates at inference" (Engdahl et al., 2026, §1) into a checkable property. Slot-level storage also allows targeted deletion of a consolidated episode.

Training and complexity. Slow parameters are meta-trained on episodes with gradients through the K within-episode writes, so CDLR is *not* BPTT-free during meta-training; at inference only $\mathbf{S}$, $\mathbf{C}$, and W change. Per token and layer, a write and a read each cost $O(nd_v)$; the workspace adds $O(Rnd_v)$ per query and the bank $O(|\{j : g_j > 0\}|\, nd_v)$. Memory is $(1 + M)nd_v b + |\mathcal{R}|nb$, constant in T and in the number of past episodes; Equation (1) applies with nd replaced by $(1 + M)nd_v$ (Figure 1).

Falsifiable hypotheses.

Hypothesis 1 (Capacity)**.** *At matched state bytes, recall accuracy of a positive, sparse, high-n Hebbian state decays with the number of stored pairs N at a rate consistent with $\tilde{O}(n)$ capacity for random keys and degrades toward $\tilde{\Omega}(\sqrt{n})$ for correlated keys (BDH Claim 7). The delta variant beats the Hebbian variant on correlated keys.*

Remark 2 (H1 after the PILOT; hypothesis unchanged). H1 is kept as preregistered. The toy pilot (Figure 3) gives three refinements to test at scale. They are not findings. (a) A *raw* positive Hebbian state with a fixed

active fraction plateaus ($N_{50} \approx 530$–600 for $n \geq 512$), because the common-mode overlap of positive keys does not shrink with n. (b) A *centred* Hebbian or *delta-rule* write with random keys gives fitted exponents of 0.86 ± 0.01 and 0.94 ± 0.01. (c) Correlated keys with a *fixed* correlation ρ as n grows, or clustered keys, give flat N_{50} (exponent ≈ 0) whenever the rule cannot remove the shared structure. This is *not* the $\tilde{\Omega}(\sqrt{n})$ regime, which assumes a different key model, so the pilot neither confirms nor refutes the worst-case half of Claim 7.

Hypothesis 2 (Dual timescale)**.** *On interleaved task streams, CDLR lowers forgetting relative to every single-state model with the same total bytes, including input-gated SSMs, at no more than a fixed relative loss in plasticity. The gain disappears when the bank is removed or its bytes are given to a larger* $\mathbf{S}$*.*

Hypothesis 3 (Workspace)**.** *Latent iterations improve accuracy on compositional demonstration-conditioned tasks over* $R = 1$ *at matched FLOPs, and halting reduces mean compute without loss in accuracy.*

Hypothesis 4 (Where synaptic memory should lose)**.** *On exact copying and high-cardinality multi-query recall at* $T < T^\star$*, softmax attention dominates every bounded-state model on accuracy and on bytes. Reporting this region is part of the result.*

5 A Matched-Compute Evaluation Protocol

Claims about new architecture classes fail most often on accounting. Cost indicators disagree (Dehghani et al., 2022), and less-tuned baselines look worse, so we fix the protocol before running it; it is frozen as a machine-readable preregistration in the supplementary material.

Principles. P1 Matched budgets. Non-embedding parameters, training tokens, and training FLOPs are matched within ±5%; every comparison is run once matched on parameters and once on FLOPs, because BDH-GPU's $O(ndL)$ per-token cost (Kosowski et al., 2025, §4.2) and $3nd$ parameter count diverge from a Transformer's. **P2 Equal tuning.** The same number of learning-rate/weight-decay sweep points per architecture; best and median of sweep reported. **P3 Modern baselines.** Transformer++ (RoPE, SwiGLU, pre-norm), Mamba or Mamba-2, GLA or Gated DeltaNet, plain linear attention at BDH's width, BDH-GPU from public code, and CDLR; xLSTM and Titans where code permits. **P4 Joint reporting.** Accuracy or loss; memory = parameter bytes + peak inference-state bytes at stated T and batch; latency p50/p95 at batch 1 and saturated throughput on named hardware; results as Pareto fronts. **P5 Statistics.** ≥ 3 seeds; Wilson intervals (Wilson, 1927); exact McNemar (McNemar, 1947) or paired bootstrap; Holm correction. **P6 Preregistration and negative results.** H1–H4, metrics, and stopping rules are frozen before runs; refuted hypotheses get the same prominence as confirmed ones. **P7 Reproducibility.** Code, configs, seeds, logs, and audit records are released (Pineau et al., 2021); closed systems are evaluated only as black boxes, and the paper says so.

E1. Language-modelling bakeoff beyond Europarl. About 30M, 125M, and 350M non-embedding parameters at ∼20 tokens per parameter (Hoffmann et al., 2022) on FineWeb-Edu (Penedo et al., 2024), with a Europarl byte-level replication (Koehn, 2005) anchoring to Kosowski et al. (2025, App. B.2). Outcomes: loss-vs-FLOPs curves (Kaplan et al., 2020), a fixed zero-shot suite, and bytes/latency at $T \in \{512, 2\text{k}, 8\text{k}, 32\text{k}\}$, which brackets $T^\star$.

E2. Associative recall, capacity, and length generalisation (Claim 7, H1, H4). Multi-query associative recall (Arora et al., 2023) with N swept until failure at $n \in \{2^{12}, \ldots, 2^{16}\}$ and matched state bytes. Keys i.i.d. and with controlled correlation. We fit $N_{50}(n)$, the N at 50% recall, and test whether its exponent is near 1 (random keys) or $\frac{1}{2}$ (correlated keys). Length generalisation: train at T, test at $2T$–$16T$ on recall, copying (Jelassi et al., 2024), and RULER-style needles (Hsieh et al., 2024), with BDH's damping ablated. Figure 3 shows a PILOT of the capacity sweep at toy width only ($n \leq 4096$, one to four octaves below the preregistered range). The pilot was used to check the analysis pipeline and to choose the key families. Preregistered E2 still needs trained encoders, MQAR, and matched-byte baselines.

E3. Ablation ladder (Figure 4). Each rung adds one component at fixed bytes and FLOPs, from plain linear attention to full CDLR. A *byte-transfer control* at rung (g) gives the bank's bytes to a larger $\mathbf{S}$,

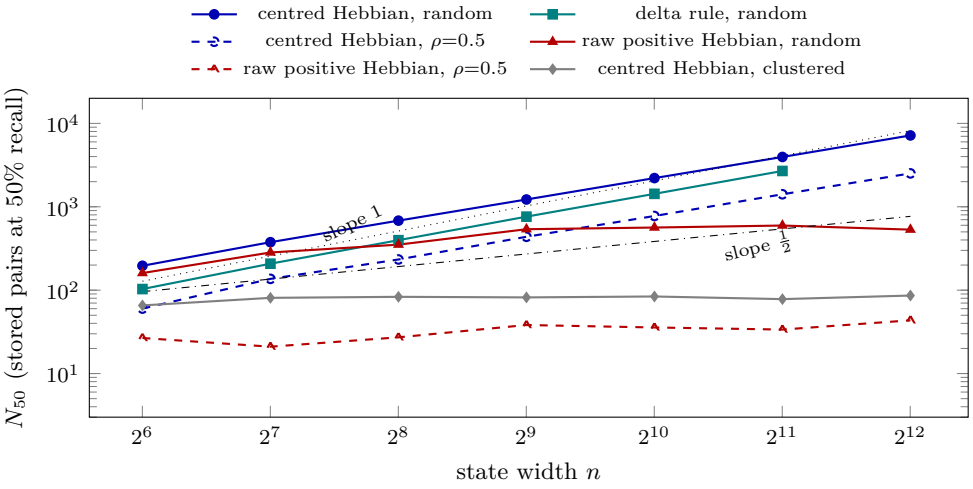


Figure 3: **PILOT (toy scale; not a test of H1 or Claim 7).** E2 capacity probe of the memory rule alone: fixed $n \times d$ state, $d = 32$, $n = 64$–4096, synthetic positive k-WTA keys (active fraction 1/16), Rademacher values, closed-set recall, 3 seeds, CPU; no trained encoder and no Transformer baseline. Points are measured; reference lines are guides. Regimes: raw positive Hebbian *plateaus* ($N_{50} \approx 550$ for $n \geq 512$). Centred Hebbian and delta rule with random keys have fitted slopes 0.86 ± 0.01 and 0.94 ± 0.01. Fixed-ρ correlated keys under the raw rule, and clustered keys, stay *flat*. This is a different regime from $\sqrt{n}$, and no slope-$\frac{1}{2}$ curve appeared. Numbers: Table 3.

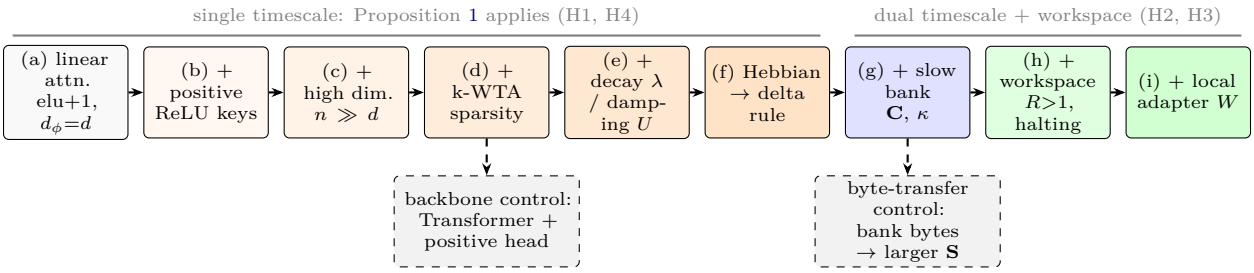


Figure 4: Conceptual E3 ablation ladder. Each rung changes one component at fixed state bytes and FLOPs; dashed boxes are controls. The diagram defines the experiment; it reports no results.

so "more memory" and "a second timescale" are not confounded; a *backbone control* replaces BDH with a Transformer plus positive head.

E4. Continual streams (H2). Streams $A \to B \to A \to \ldots$ with controlled interference: permuted mappings, class-incremental symbol tasks, and language-domain streams. Metrics: average accuracy, forgetting, backward and forward transfer (Lopez-Paz & Ranzato, 2017), relearning speed, plasticity loss (Dohare et al., 2024), and total bytes. Baselines: frozen, full fine-tuning, replay (Rolnick et al., 2019), EWC (Kirkpatrick et al., 2017), a fast-state-only model, and input-gated SSMs, each with and without task-boundary signals.

E5. Demonstration-conditioned latent reasoning (H3). ARC-AGI-1 public evaluation, ConceptARC (Moskvichev et al., 2023), and RE-ARC controls (Hodel, 2024), with no evaluation tasks or task identifiers in training (Engdahl et al., 2026, §1). BDH-CQ appears only as (i) its *reported* numbers or a black-box run if access is granted, or (ii) a *clearly labelled reimplementation* of its Eqs. 1–4 ("BDH-CQ-R"). Contamination controls use disjoint hashes and a freshly authored held-out set; cost per task is reported under an explicit accounting (GPU type, seconds, price, pipeline stages).

Table 2: Claims, evidence, and gaps. [P]: paper-stated limitation; [I]: our inference.

Claim (source)	Evidence reported	Gap
BDH rivals GPT-2 at 10M–1B params (Kosowski et al., 2025, Abstract; §4.2, Fig. 7)	Europarl En–Pl/En–Cs byte-level, 1.2B tokens; GPTXL baseline (nanoGPT + Transformer-XL buffer + ALiBi), App. B.2, Table 4	[I] One corpus family; values graphical only; no modern linear/SSM baselines in experiments though cited (§4.3); no FLOP-matched curve.
Monosemantic synapses (§6.3)	Mann–Whitney $U = 2368$, $p < 10^{-14}$, 50+50 LLM-generated sentences, one concept pair, one layer	[I] Single concept pair; no comparison with SAE features in Transformers (Bricken et al., 2023).
Capacity $\tilde{O}(n)$ / $\tilde{\Omega}(\sqrt{n})$ (Claim 7, §6.1, App. C.2)	Proof sketch (JL + Azuma)	[I] Not measured on recall benchmarks; motivates E2.
Reasoning generalises over length/time (§1)	–	[P] Thermodynamic-limit/PAC-like bounds are a goal (Abstract, §1). [I] Not directly tested.
Long context, no hard bound (§6.1)	–	[P] Damping (RoPE+ALiBi-like) still required.
Model merging by concatenation in n (§7.1, Table 2)	Merged 38M model, X→En loss 0.39–0.43	[P] Merged model mixes languages when translating from English.
Training without BPTT (§7.2)	English loss ≈ 0.65 → 0.75–1.05	[P] Degradation; cross-language concept matching lost.
Sparse/neuromorphic execution (§2.4)	Cost estimate	[P] CPU sparse kernels 10^2–$10^5\times$ less cost-effective than GPU BDH-GPU. [I] Energy not measured.
BDH-CQ 29.5% pass@2 ARC-AGI-1, $0.00070/task (Engdahl et al., 2026, Abstract; §5, Table 1)	118/400, Wilson [25.24, 34.15]; ≈0.85 H200 GPU-s at $3/h; independent black-box audit	[I] §6.6 prices STANDARD effort (also 118/400) at $0.00265246/task and MIN at $0.00088399; relation to the $0.00070 headline not stated.
Architecture	"Dimensions, exact update rules, and implementation details remain proprietary" (§3.3); recipe proprietary (§4.1)	[P] Not reproducible from text. [I] No ablation separating BDH backbone from candidate construction/ranking.
Generalisation to ConceptARC (§6.1, Table 1)	95/160 semantic, 96/160 opaque pass@2	[P] ConceptARC is in the training mixture (§4.2); exposure via training or checkpoint selection not ruled out (§6.5).
pass@2 semantics (§6.6)	75 single-candidate records	[P] pass@2 is not two independent attempts for every input.
Compositional boundaries (§6.2–6.3, App. A.3)	Ordering 1/24 at length 8; colour swap 0/72 composed; conditional rule selection 100%→56.7%	[P] Stated weaknesses. Ladders are single-family, 40 tasks (App. A.4); mechanic rates confounded by LLM authoring (App. A.2).
Scaling "1B to 600B" (§9.2)	No figure, table, or numbers	[I] Unsupported by reported data.

Compute envelope. At $6ND$ FLOPs the E1 grid (30M/0.6B, 125M/2.5B, 350M/7B tokens) costs $\approx 1.7 \times 10^{19}$ FLOPs per architecture-seed, or $\approx 2.5 \times 10^{20}$ for five architectures and three seeds: about 550 A100-hours at 40% utilisation (≈1.1k with sweeps). E2–E4 are an order of magnitude cheaper.

6 Critical Analysis of BDH and BDH-CQ

Both papers make real contributions. BDH gives an explicit bridge from attention to local synaptic dynamics, with a clear parameter count, public code, and testable theory. BDH-CQ reports an independent black-box audit that reproduced its 29.5% pass@2 without weight access (Engdahl et al., 2026, §5), byte-identical repeat runs (§6.6), Wilson intervals, and candid failure analyses (§6.2–6.4). Our critique concerns what the evidence licenses (Table 2).

Open problems. (1) *Measured capacity*: does BDH-GPU's high-n positive state reach Claim 7's regime, and beat gated/delta models at equal *bytes*? (E2; the toy PILOT in Figure 3 suggests that a raw positive state needs centring or an error-correcting write to scale at all. This is untested with trained encoders.) (2) *Which ingredient matters*: positivity, sparsity, high n, weight sharing, or damping? No published experiment isolates these. (E3) (3) *Interference*: merging by concatenation and continual adaptation both need a mechanism that keeps tasks apart, which decay alone cannot provide (Proposition 1; E4). (4) *Latent reasoning without a proprietary pipeline*: how much ARC performance survives an open reimplementation of Eqs. 1–4? (E5) (5) *Cost accounting*: cost-per-task frontiers need one published accounting of hardware, pipeline stages, and effort tier.

7 Discussion

Why a dual timescale. Softmax attention avoids the plasticity–stability dilemma by never forgetting within the window and never learning across windows. Fixed-state models learn online but share one state, so by Proposition 1 they trade retention against interference. Weight-level continual learning (Kirkpatrick et al., 2017; Zenke et al., 2017) works on the slowest timescale and needs gradients. CDLR adds the missing inference-time timescale, by analogy with hippocampal-to-neocortical transfer (McClelland et al., 1995) and multi-timescale synapses (Benna & Fusi, 2016), while memory stays bounded and slow weights stay frozen. H2 is falsifiable, and E3's byte-transfer control is designed to refute it if it is wrong.

Where CDLR should lose. We hold H4 as likely: below $T^\star$, and on exact copying or high-cardinality recall (Arora et al., 2023; Jelassi et al., 2024), a KV-cache is both smaller and more accurate. A hybrid of local softmax attention plus CDLR memory is the natural compromise and belongs in E1–E2. Bank selection needs a task-boundary/novelty signal or a labelled support set, and the unsupervised gate may fail where tasks share inputs.

Limitations. (i) No large-scale results are reported; the protocol's cost is stated above. (ii) Proposition 1 is a linear, expected-value statement; nonlinear readouts and input-dependent gates need a richer analysis. (iii) CDLR adds hyperparameters $(M, \tau, \beta, \gamma, R, c)$ that could absorb tuning budget, hence P2. (iv) Meta-training uses BPTT across demonstrations. (v) The BDH-CQ audit relies on its text alone, because the system is closed. (vi) Positive sparse codes and slot-structured memory give audit signals (which neurons wrote, which slot was read, how many steps ran), not explanations. (vii) The E2 PILOT is a regime characterisation at toy width ($n \leq 4096$, synthetic keys). It does not extrapolate to $n = 32{,}768$, to trained encoders, or to matched-compute comparisons with Transformers.

8 Conclusion

Softmax attention, Hebbian synaptic state, and recurrent latent memory are three settings of one associative-memory operator, differing in growth, key geometry, forgetting, consolidation, and workspace. This view yields a closed-form memory crossover, exposes retention–interference coupling in any single-timescale state, and motivates CDLR, a fast episodic state plus a slow gated bank with a latent workspace and an audit trail. We separated what BDH and BDH-CQ state from what we infer, and specified a matched-compute protocol under which they and CDLR can be confirmed or refuted, including the cases where the KV-cache wins.

Broader Impact Statement

Models that learn at inference change behaviour after deployment. This raises the risk of memory poisoning through crafted demonstrations, by analogy with training-time poisoning (Wallace et al., 2021), and the risk that consolidated slots retain personal data. The CDLR audit log, parameter hash, and slot-level deletion are partial mitigations that make writes visible and reversible; deployments should cap consolidation rates, require provenance for consolidated episodes, and support rollback. Bounded-memory inference can lower serving energy for long contexts, but only above $T^\star$, and this should be measured, not assumed.

Code availability. An anonymized archive with the protocol code, configurations, the frozen preregistration (`prereg/e1_e5_protocol_v0.json`), smoke-test logs, and the E2 PILOT script and raw results (`e2_capacity/`) is provided as supplementary material; a public repository link will be added in the camera-ready version.

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A Preliminary Smoke Tests and the E2 PILOT (Not Main Results)

Status. Single-machine CPU runs on synthetic toy tasks with 5×10^4 to 1.6×10^6 parameters. They check that the protocol and audit pipeline run end to end and illustrate what mismatched budgets do. They provide *no* evidence about the hypotheses at scale, about ARC, or about the BDH §7.2 gap. Numbers come from run files in the supplementary archive (seed and date in the logs).

P1: Episodic write without weight change (Tiny-CDLR, 52,531 params). BDH backbone (public `bdh.py`, unchanged, n_{embd}=32, 2 layers, 256 sparse units), K=4 demonstrations per episode on a random key→value map with 16 classes (chance 0.0625). Accuracy with demonstrations written into $\mathbf{S}$ was 1.00, vs. 0.086 for the same queries with an empty $\mathbf{S}$. The parameter hash was identical before and after inference. Mean halting step was 1.23 of R=3.

P2: A→B→A with full interference. A and B are different permutations of the same 16 inputs. A fast-state-only arm forgot A completely after B (A accuracy 1.00 → 0.00; first-5 relearning accuracy 0.04). The dual-timescale arm kept A at 1.00 (first-5 relearning 0.80). *Caveat:* slot selection used a labelled 32-token support set at evaluation, which is a task-identity signal, so this probe does not test the unsupervised gate.

P3: Local adapter vs. BPTT (negative result). On a lag-1 permutation stream over a small pretrained BDH (3 seeds, 150 steps/phase), a local delta-rule readout with fast Oja memory reached 0.81 average online accuracy vs. 0.91 for full BPTT, at about 2.3× lower per-step cost. *Both forgot task A almost completely* (forgetting gap 0.97 in both), as Proposition 1 predicts for a single shared state.

P4: Why matched budgets matter. A 300-step Shakespeare character run had BDH reach validation loss 1.875 vs. 2.43–2.44 for a Transformer and a linear-attention model, but BDH had 1.9× the parameters (1.64M vs. 0.87M), took 12× the wall time, and the baselines were still on an early plateau: the comparison is not matched and supports no ranking. In a toy Tiny-CDLR vs. Transformer recall comparison, the undertrained Transformer was far less accurate (0.25 vs. 1.00) but 28× faster and used less memory at this context length.

P5 [PILOT]: Toy capacity sweep (E2; Figure 3). We used a fixed $n \times d$ state ($d = 32$, $n = 64$–4096) with write $\mathbf{S} = \sum_i \phi(k_i)v_i^\top$ and read $q^\top \mathbf{S}/q^\top z$ ($\lambda = \eta = 1$, no decay). There were three rules. *raw* is the positive outer product. *centred* subtracts the batch-mean key before write and read. *delta* is a sequential normalised delta rule ($n \le 2048$, $N \le 16n$). Keys were positive and k-WTA sparse (active fraction 1/16): i.i.d. (*random*), $\sqrt{\rho}\,\text{shared} + \sqrt{1-\rho}\,\text{private}$ (*corr* ρ, with ρ *fixed* as n grows), or 8 centres plus noise (*clustered*). Values were Rademacher in $\mathbb{R}^{32}$, and recall was argmax over the N stored values. N was swept on a quarter-octave grid. N_{50} is log-linearly interpolated, and the slope is a least-squares fit of $\log N_{50}$ on $\log n$ (3 seeds; about 41 s on 8 CPU threads). *Reading (regime characterisation only).* Every curve has an exponent

Table 3: P5 [PILOT] N_{50} (mean over 3 seeds) and fitted exponent (± s.e.). "–": not run (delta at $n = 4096$). Where noted, delta n=2048 uses 2/3 seeds.

rule	keys	n=64	256	1024	2048	4096	exponent
raw Hebbian	random	160	351	564	596	532	0.29 ± 0.07 (plateau)
raw Hebbian	corr 0.5	27	27	36	34	43	0.14 ± 0.04 (flat)
centred	random	196	682	2208	3958	7172	$\mathbf{0.86 \pm 0.01}$
centred	corr 0.2	109	420	1267	2317	4148	0.86 ± 0.02
centred	corr 0.5	60	233	775	1412	2514	0.88 ± 0.02
centred	corr 0.8	42	159	511	905	1626	0.88 ± 0.01
centred	clustered	66	83	84	78	86	0.04 ± 0.02 (flat)
delta	random	103	398	1430	2688	–	$\mathbf{0.94 \pm 0.01}$
delta	corr 0.5	25	112	472	846	–	1.01 ± 0.03
delta	corr 0.8	5	5	6	4	–	-0.02 ± 0.06 (flat)
delta	clustered	36	43	37	35	–	0.05 ± 0.08 (flat)

near 1, or ≤ 0.3. Which one appears depends on whether the write rule can remove the key-correlation structure. A single shared direction is removed by centring, at a constant-factor cost of 2–4.5×. Several shared directions (clustered keys) are not removed, and neither is $\rho = 0.8$ under the delta rule. Because ρ is held fixed as n grows, the flat curves are a different regime from the $\tilde{\Omega}(\sqrt{n})$ worst case, and they do not refute it. The largest widths reach $N_{50}/n = 1.75$ (centred, $n = 4096$) and 1.31 (delta, $n = 2048$). These figures come from toy width, synthetic keys, and no encoder, and they do not establish general capacity at any scale.

B Reporting Checklist for Architecture Bakeoffs

1. Non-embedding params, training tokens, training FLOPs (method stated), wall time, hardware.
2. Sweep grid and size per architecture; best and median of sweep.
3. Inference bytes: parameters + state at each reported T and batch; latency p50/p95; throughput.
4. Seeds, intervals, paired tests, multiple-comparison correction.
5. Data provenance and contamination checks (hash overlap) for every evaluation set.
6. For closed systems: black-box status, access date, and exactly which numbers are quoted.
7. Preregistered hypotheses with outcome (confirmed / refuted / inconclusive).

C Notation

T	sequence length (tokens)
n	neuron dimension (BDH) / key-feature dimension
d, d_v, d_{model}	low-rank width, value width, Transformer width
$\mathbf{S}, z$	fast episodic state and normaliser
$\mathbf{C}_j, \mathbf{k}_j, M$	slow bank slot, slot key, slot budget
λ, η	fast decay, fast write rate
κ, β, τ	consolidation map, merge rate, merge threshold
$\mathbf{H}_r, R, c$	workspace state, max steps, halting confidence
$T^\star$	byte crossover between KV-cache and bounded state (Equation (1))